LEC 4.1: GENERAL MECHANISMS OF CELL SIGNALING

Learning goals:

- To understand cell signaling/communication characteristics
- To learn about the types of cell communication
- To know how cells communicate with other cells (via signaling molecules or direct contact) or to ECM components

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CELL SIGNALING

Communication of cells for the regulation of tissue development, differentiation, homeostasis and adaptation to insults (from embryogenesis to adulthood)

Coordination of cells and tissues is dependent on the integration of signals received by each cell to elicit a response

Long-distance (distal) communication





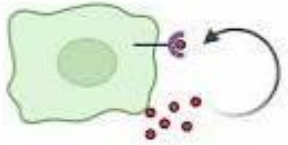
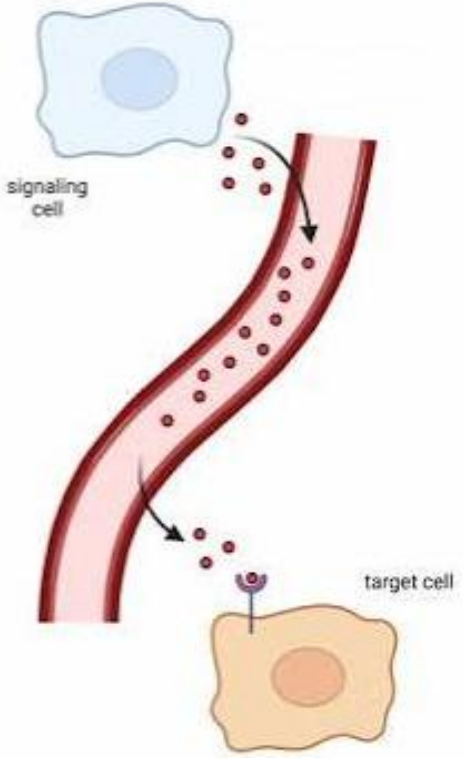
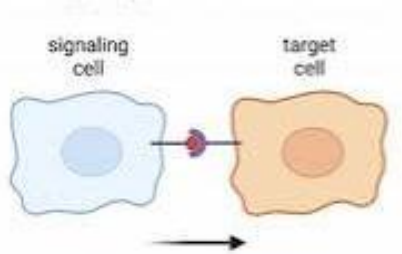
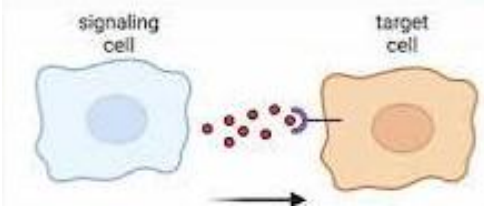
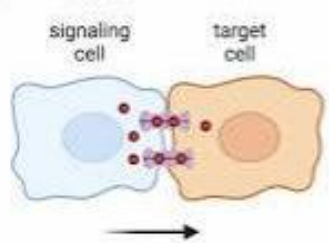
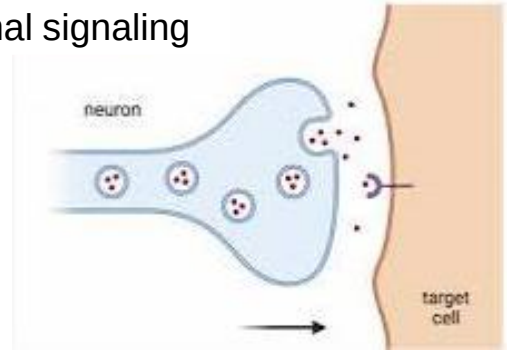
Close (local) communication







TYPES OF CELL COMMUNICATION

CHEMICAL SIGNALING		DIRECT SIGNALING
Autocrine signaling 	Endocrine signaling 	Juxtacrine signaling 
Paracrine signaling 		Signaling through gap junctions 
Neuronal signaling 		

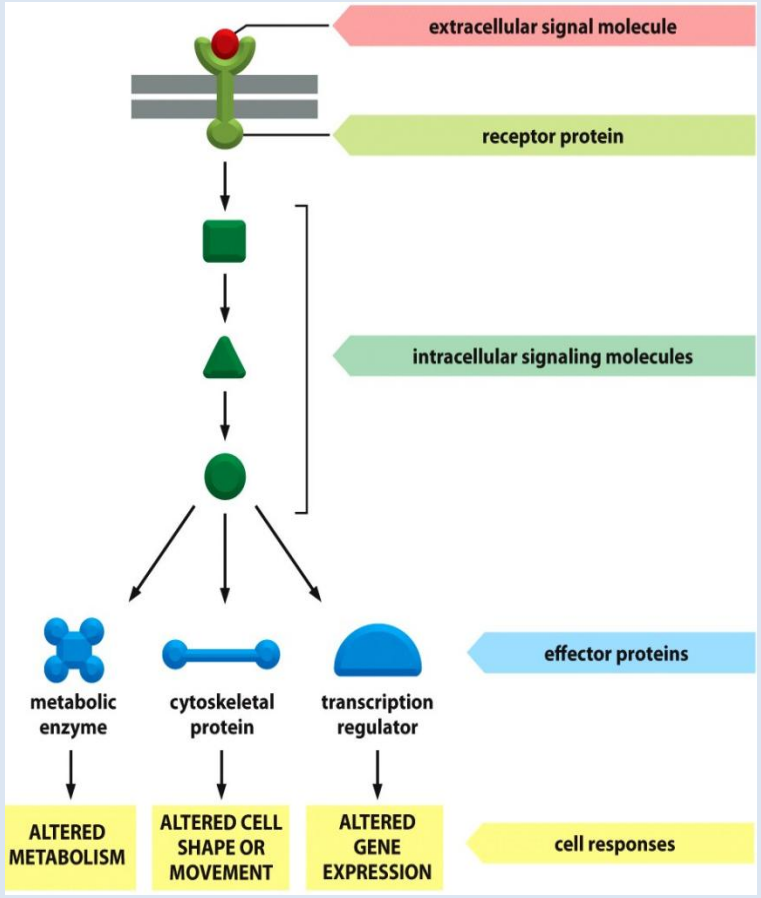
CELL SIGNALING: CHARACTERISTICS

1. **Response through specific receptors** (plasma membrane or cytoplasmic) **or target enzymes**

2. **Intracellular signaling cascade** (i.e., protein activation and gene transcription)

SIGNALING THROUGH HYDROPHILIC LIGANDS

- **Hydrophilic** ligands (bind extracelular receptors)
- Receptor proteins in plasma membrane (PM)
- Intracellular signaling/transduction cascade
- Effector proteins: cytoplasmic proteins, transcription factors (activate response genes)
- Cellular response



CELL SIGNALING: CHARACTERISTICS

1. **Response through specific receptors** (plasma membrane or cytoplasmic) **or target enzymes**

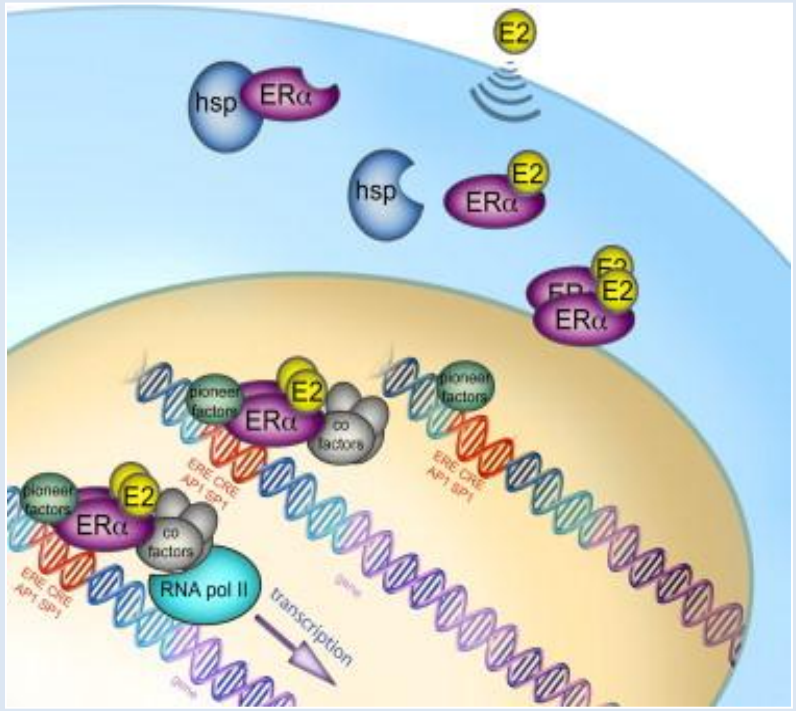
2. **Intracellular signaling cascade** (i.e., protein activation and gene transcription)

SIGNALING THROUGH HYDROPHOBIC MOLECULES

- **Hydrophobic molecules** (diffuse through PM)
- Effector proteins:

Intracellular receptor proteins function as transcription factors, or cytoplasmic enzymes produce secondary messengers

- Cellular response



CELL SIGNALING: CHARACTERISTICS

1. Response through specific receptors (plasma membrane or cytoplasmic) **or target enzymes**

2. Intracellular signaling cascade (i.e., protein activation and gene transcription)

3. On/off system: inactivation of signaling cascade by:
- Receptor mediated endocytosis/pinocytosis
- Phosphatases (inactivate proteins activated by kinases)
- GTP dissociation to GDP + Pi
(GTPase Activating Proteins, GAPs)

4. Integration of simultaneous signals → signaling output

5. Amplification of the signaling process
(1 rhodopsin molecule → 500 phospho-diesterases → 10^5 cGMP)

6. One signaling molecule can have different effects:
Acetylcholine (myocardiocytes, smooth muscle cells and salivary glands)

1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

TYPES OF SECRETED MOLECULES

A. HYDROPHOBIC MOLECULES

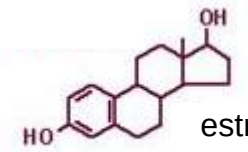
Cholesterol

- STEROID HORMONES (lipid hormones)
  - Sexual esteroids
    - Estrogens and androgens
  - Corticosteroids
    - Glucocorticoids: stimulate glucose production
    - Mineralocorticoids: Regulate saline and hydric balance
- VITAMIN D_3: calcium metabolism and bone growth

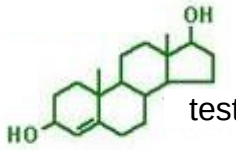
- NITRIC OXIDE: hypertension and erectyle dysfunction

Arachidonic acid

- EUCOSANOIDS
  - Prostaglandins
  - Tromboxanes
  - Leukotrienes
    - Regulation of vascular and bronchial tone, or inflammation

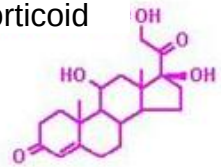


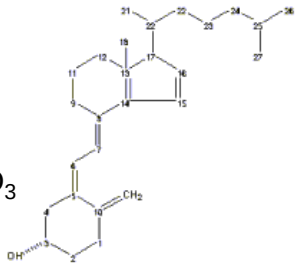
estradiol



testosterone

glucocorticoid




Vit D_3

1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

TYPES OF SECRETED MOLECULES

B. HYDROPHILIC MOLECULES

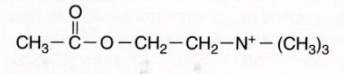
- NEUROTRANSMITERS: Acetylcholine, Dopamine, Adrenaline, Histamine, Serotonine, Glutamate, Gamma-aminobutyric acid (GABA)

- PEPTIDE HORMONES: Insuline, Glucagon, Prolactine, Growth Hormone (GH), Follicule Stimulating, Hormone (FSH)

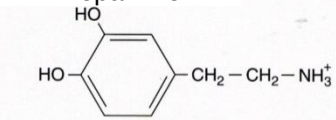
- NEUROPEPTIDES: Enkephalins and Endorphins (natural pain killers)

- GROWH FACTORS: Epidermal Growth Factor (EGF), Plateled-derived growth factor (PDGF), Transforming Growth Factor (TGF), Vascular Endothelial Factor (VEGF)

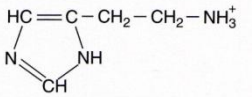
Acetylcholine

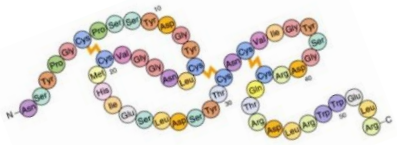


Dopamine



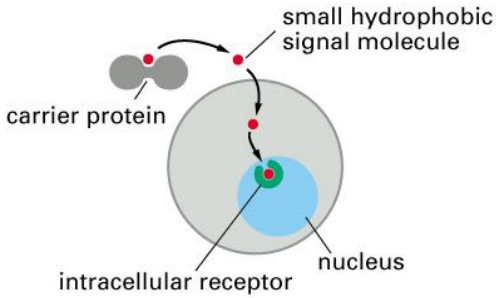
Histamine





1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

A. HYDROPHOBIC MOLECULE RECEPTORS

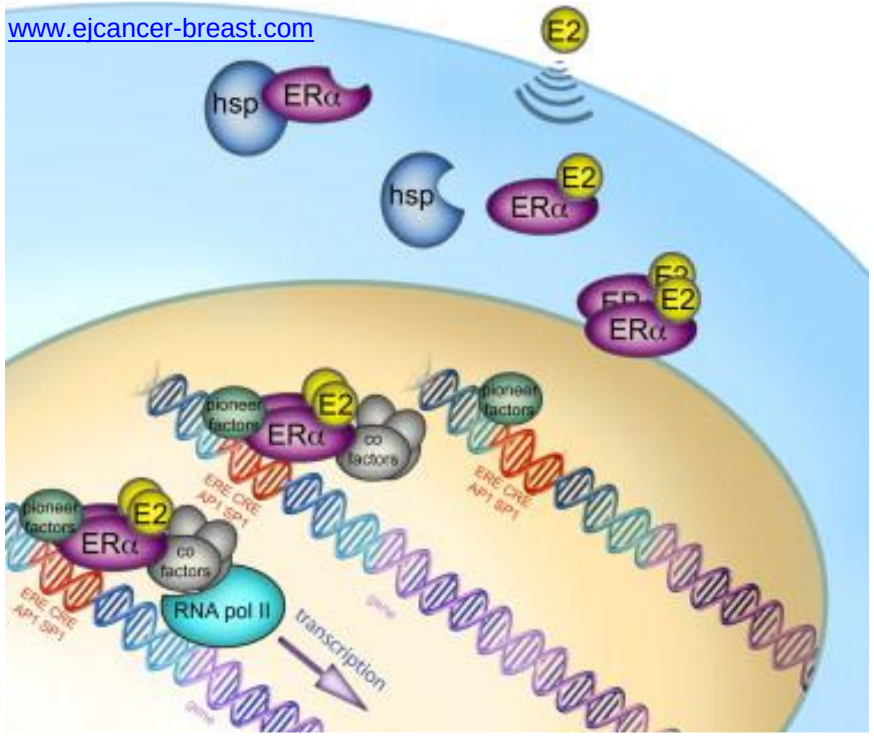


A.1. Example of HORMONE SIGNALING: Esteroid receptors super-family (example: the estrogen receptor)

- Intracellular (cytoplasmic) receptor

- Receptor dimerization upon ligand binding

- Dimer enters the nucleus and functions as a transcription factor (to control gene expression)



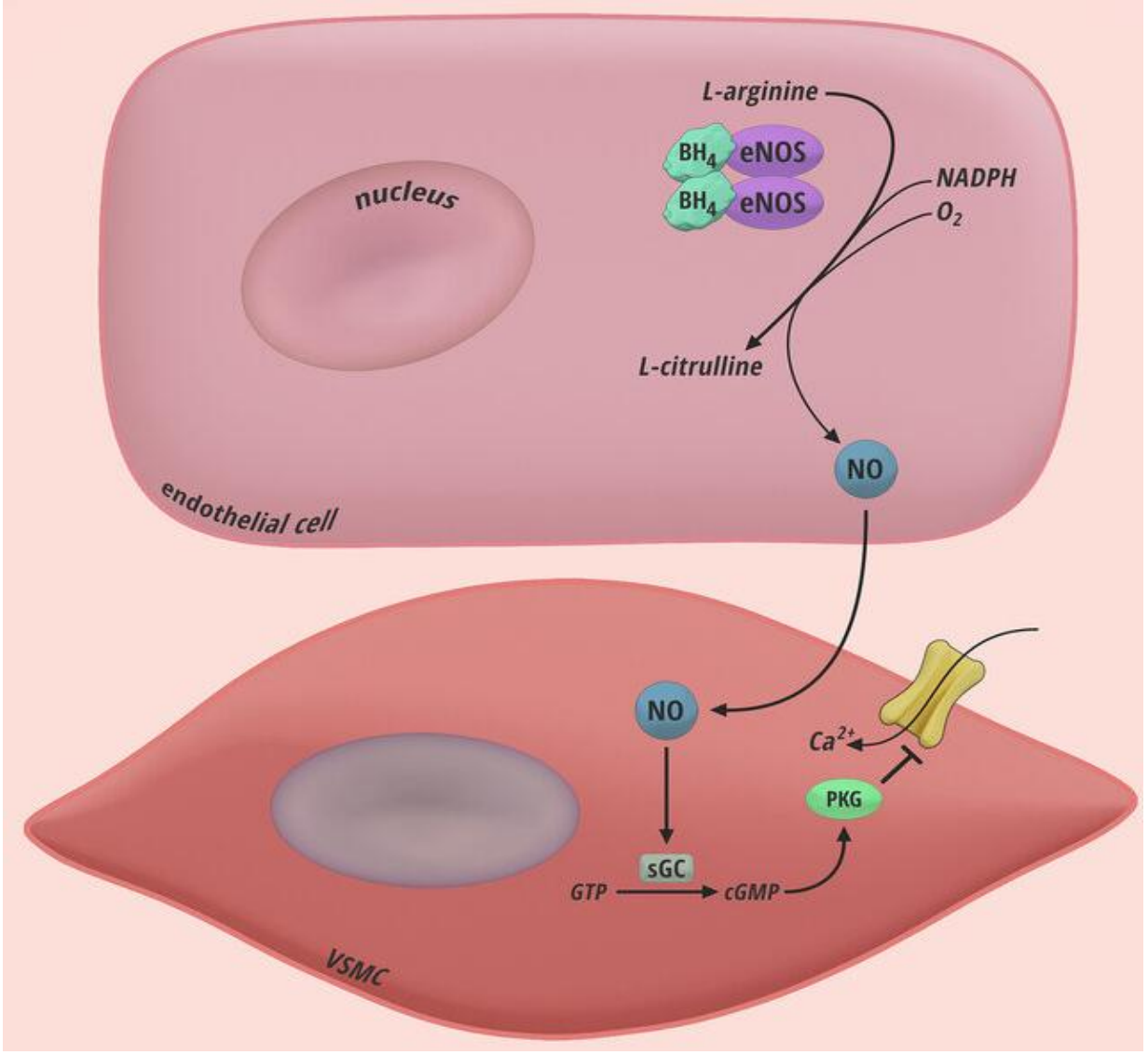
1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

A. HYDROPHOBIC MOLECULE RECEPTORS

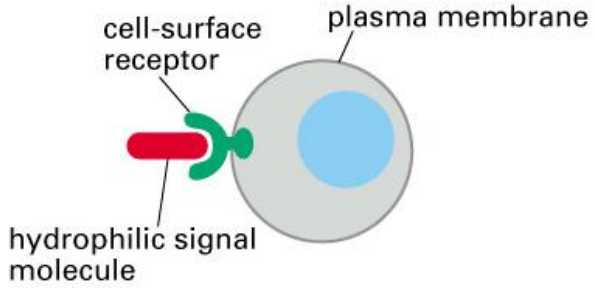
A.2. Example of GAS SIGNALING: Nitric oxide (NO)

Regulates blood vessel contraction by decreasing muscle tension
(Drs. Furchgott, Ignarro and Murad: Nobel Prize in Physiology-Medicine 1998)

Angina and chronic heart failure
Nitroglycerine or aspirin as NO donor to relax smooth muscle and dilate blood vessels

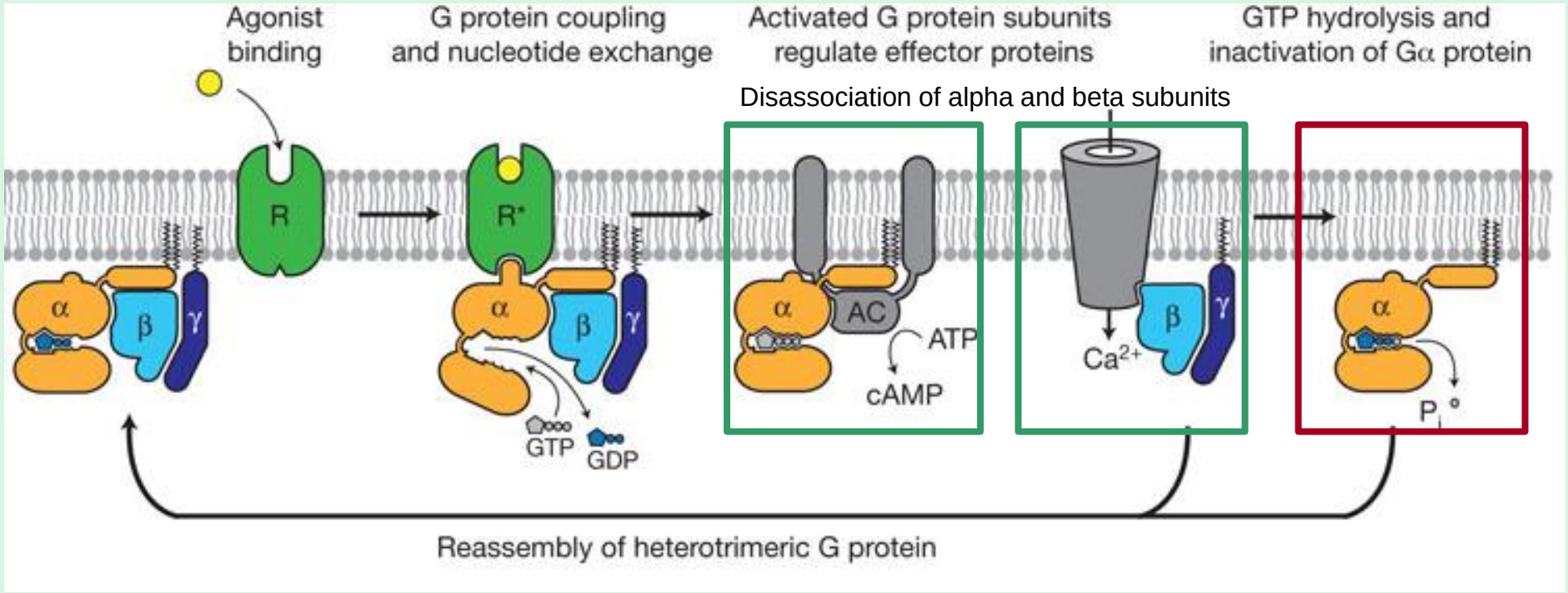


1. CELL COMMUNICATION THROUGH SECRETED MOLECULES



B. HYDROPHILIC MOLECULE RECEPTORS

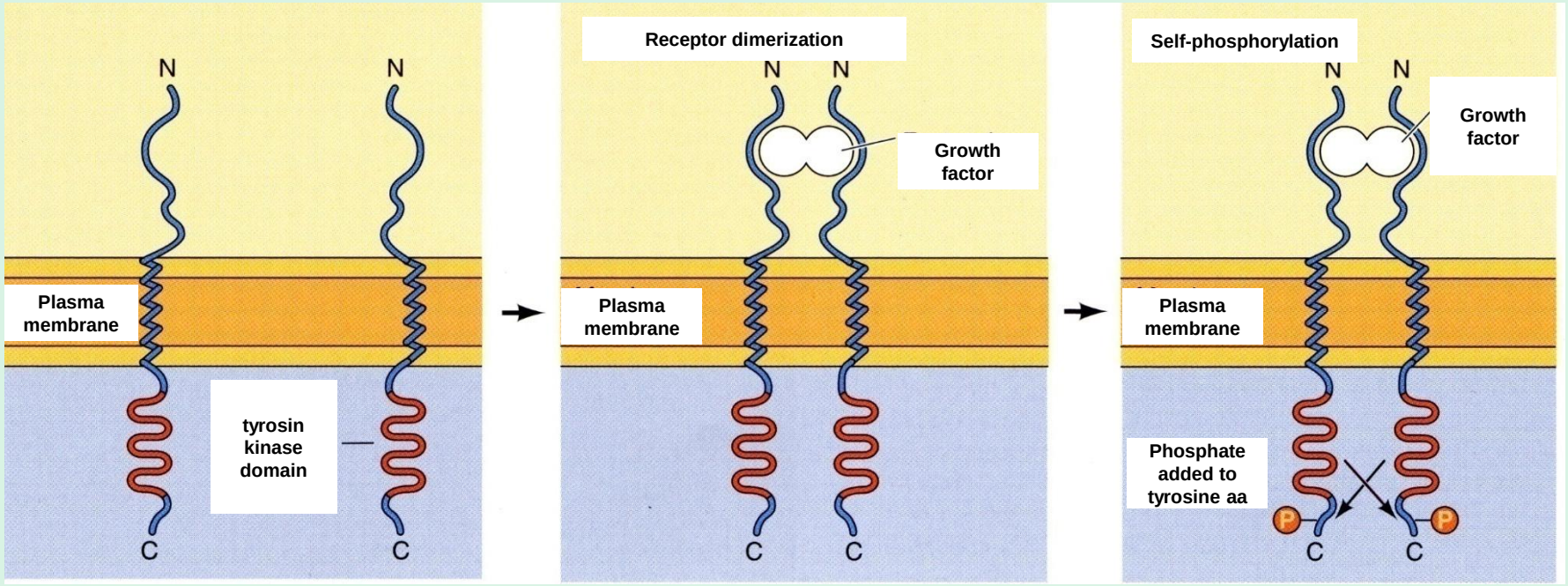
B.1. Signaling through G protein-coupled receptors



1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

B. HYDROPHILIC MOLECULE RECEPTORS

B.1. Signaling through catalytic receptors (Tyr Kinase -TK- and Ser/Thr Kinase -STK- receptors)



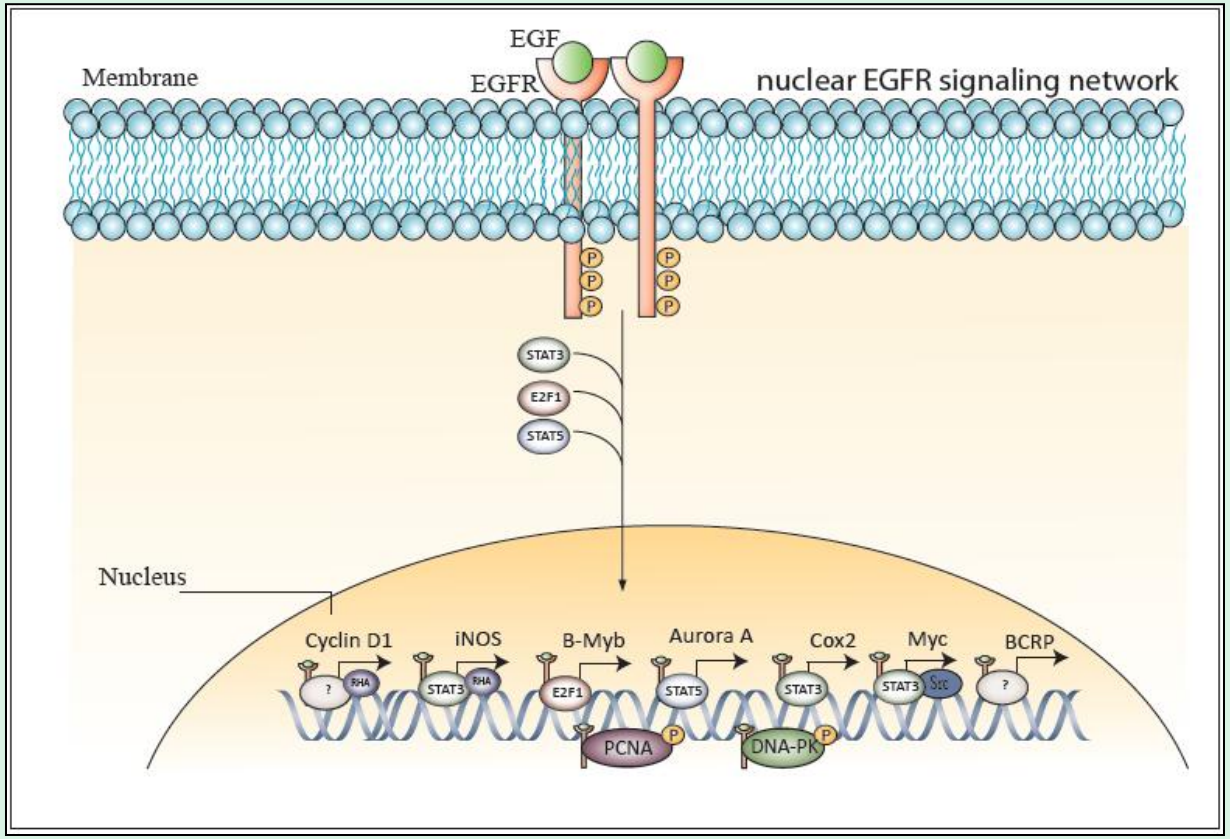
Examples of TK receptors: EGFR, PDGFR, TrkR (NGF), TGFβR, IGF1R, FGFR, VEGFR

1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

B. HYDROPHILIC MOLECULE RECEPTORS

B.1. Signaling through catalytic receptors (Tyr Kinase -TK- and Ser/Thr Kinase receptors)

Phosphorylation of transcription factors

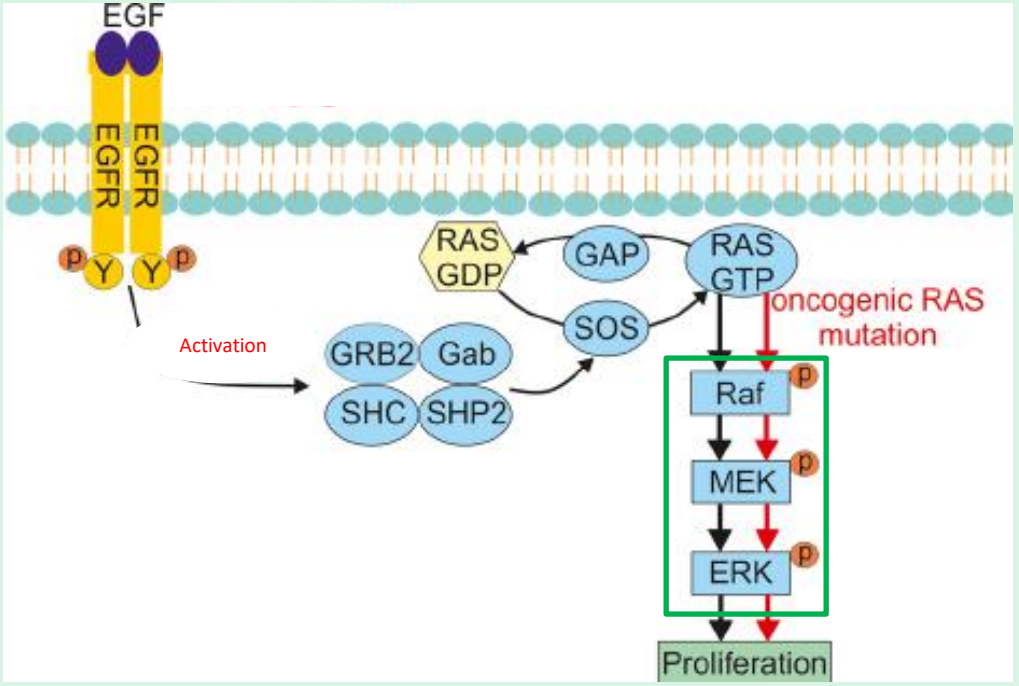


1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

B. HYDROPHILIC MOLECULE RECEPTORS

B.1. Signaling through catalytic receptors (Tyr Kinase -TK- and Ser/Thr Kinase receptors)

Activation of the RAF-MEK-MAPK module

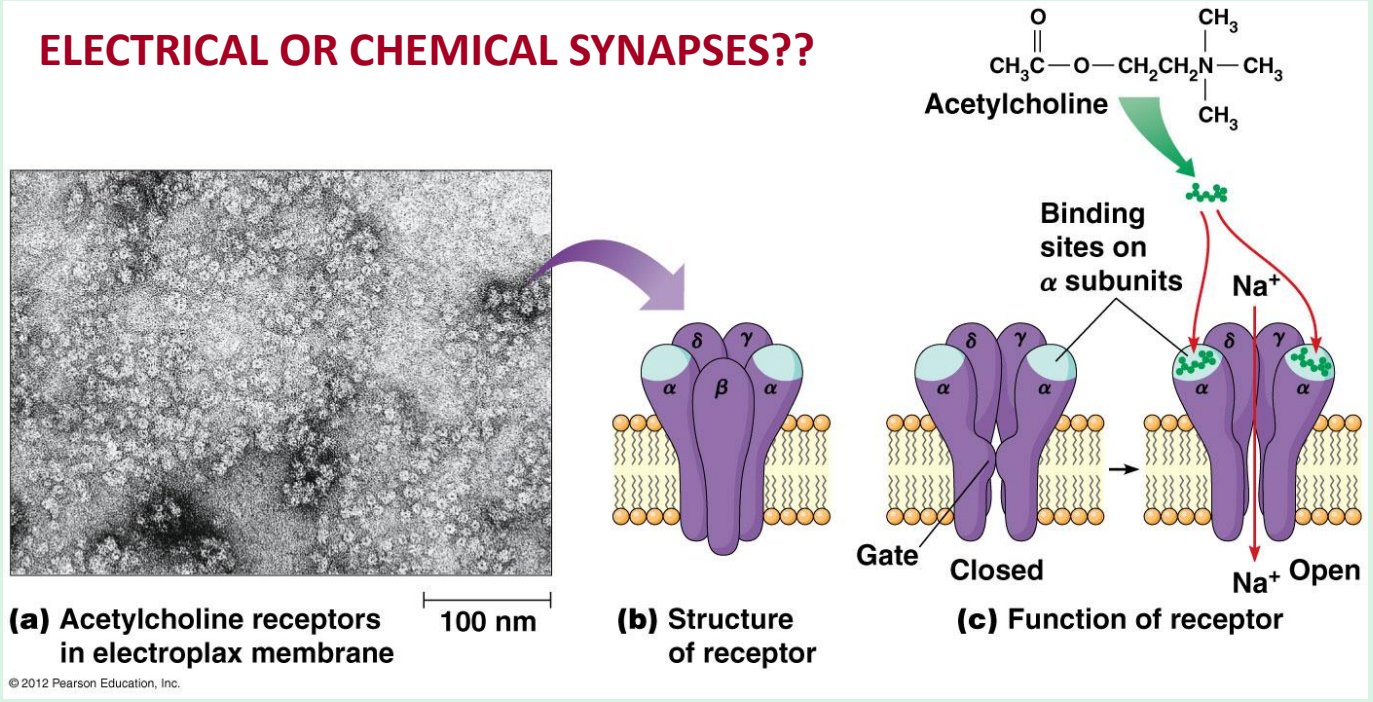


1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

B. HYDROPHILIC MOLECULE RECEPTORS

B.2. Signaling through ionic channels

Neurotransmiters

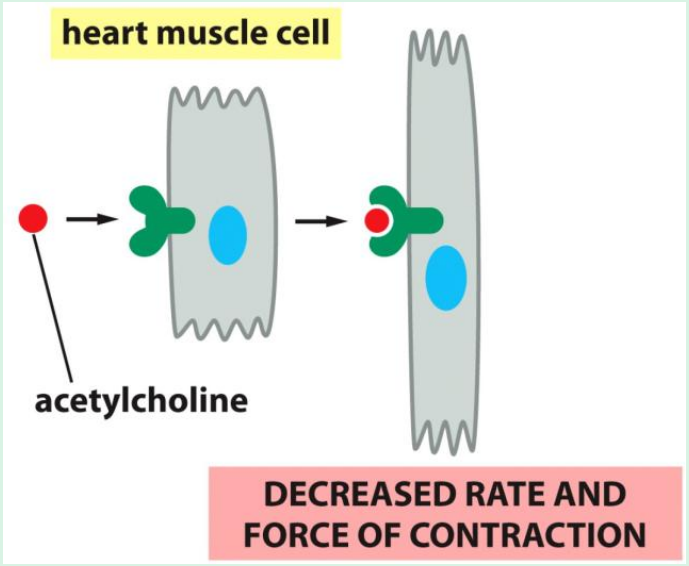


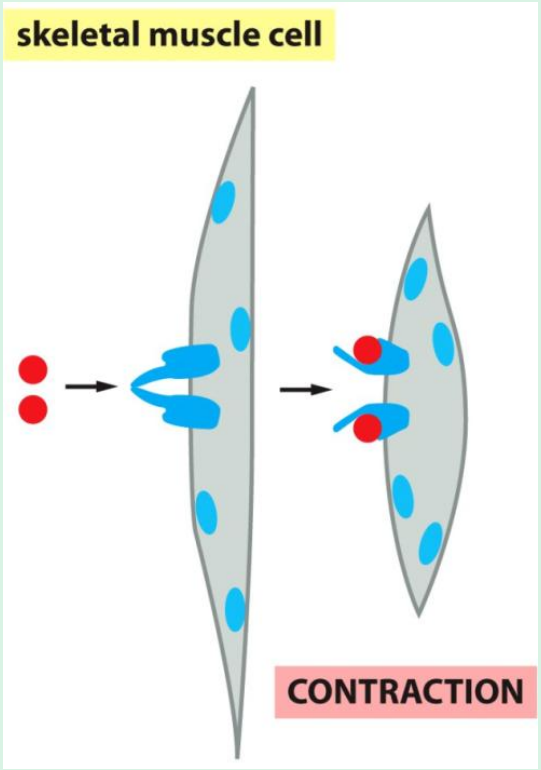
1. CELL COMMUNICATION THROUGH SECRETED MOLECULES

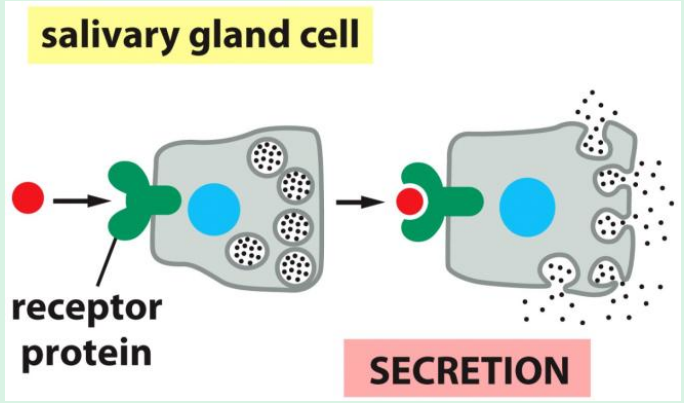
B. HYDROPHILIC MOLECULE RECEPTORS

B.2. Signaling through ionic channels

Neurotransmiters





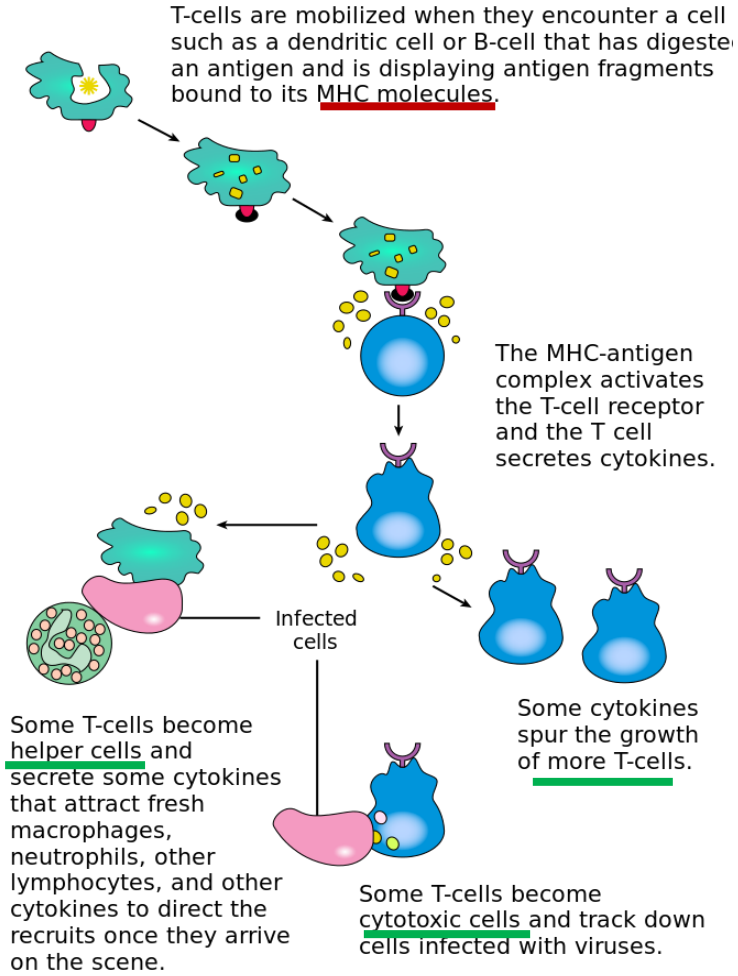


2. CELL-TO-CELL COMMUNICATION

Contact of membrane proteins of **one cell to another cell**

A. Example: activation of T-lymphocytes

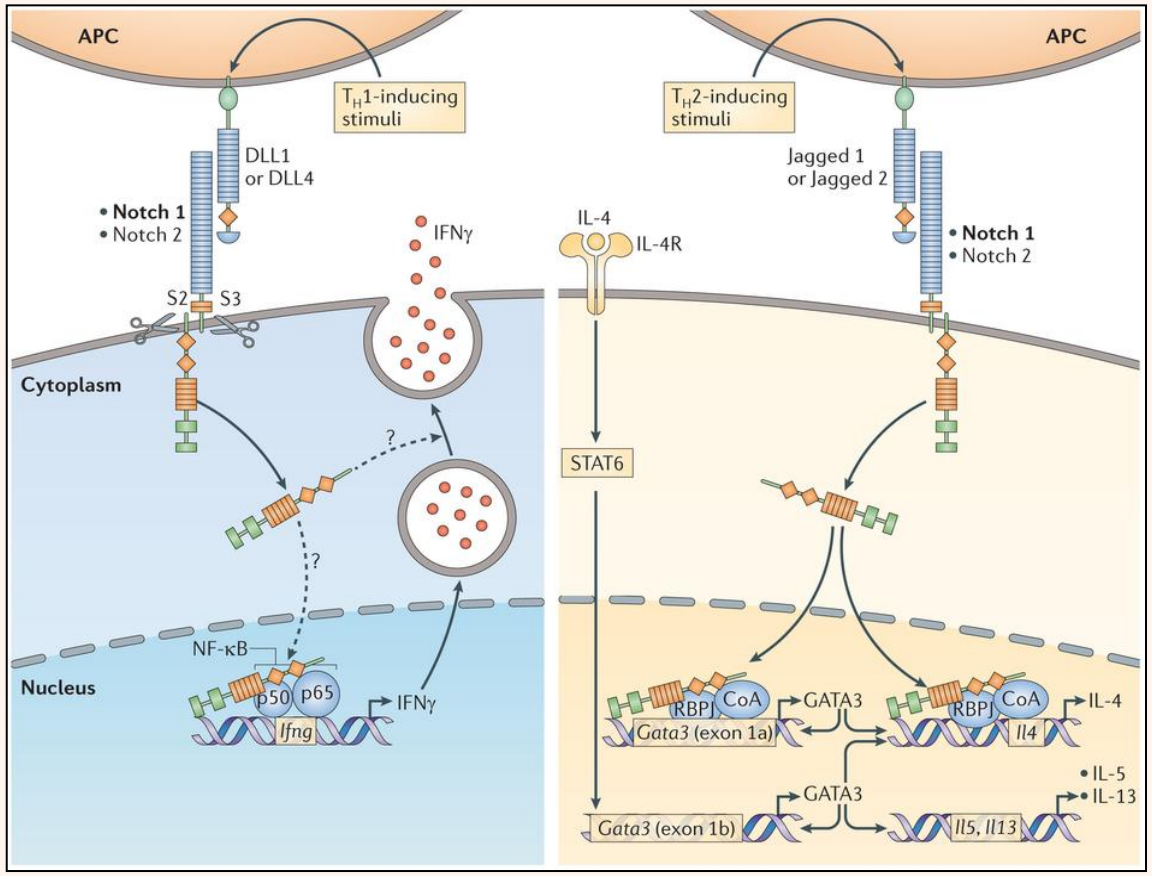
- Antigen presenting cells (dendritic cells/B-cells)
- **Antigen** presented to T-lymphocytes through **MHC molecules**
- Additional contact of membrane proteins
- Both signals activate T-lymphocytes (IL2)
- T-lymphocyte secretes cytokines:
  - Autocrine signal: cell proliferation and cytotoxicity
  - Paracrine signal: chemotractant activity (macrophages and neutrophils



2. CELL-TO-CELL COMMUNICATION

Contact of membrane proteins of **one cell to another cell**

B. Example: Notch receptor (1-4) and ligands (delta, jagged and serratum)



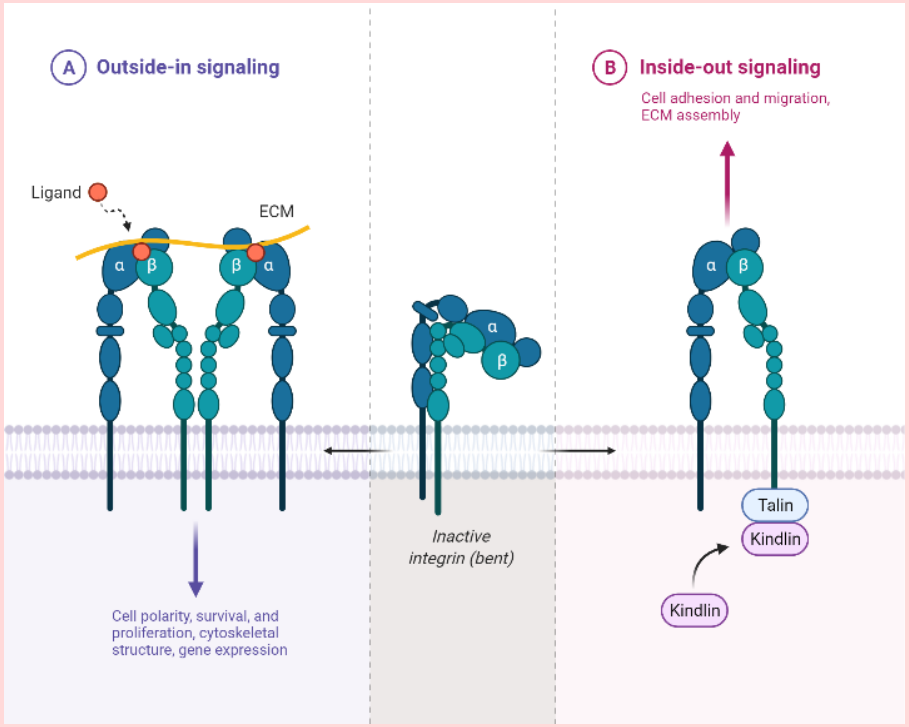
- Embryonic development (neuron differentiation)

- Tumorigenesis (cancer stem cells)

3. CELL-TO-ECM COMMUNICATION

Contact of membrane proteins of **one cell to the extracellular matrix**

Example: integrin-laminin or integrin-fibronectin binding



- Cell differentiation into tissues (e.g., epithelial cells)
- Proliferation (e.g., Sertoli cells)
- Migration (e.g., neural crest cells)

4. GAP/COMMUNICATING JUNCTIONS

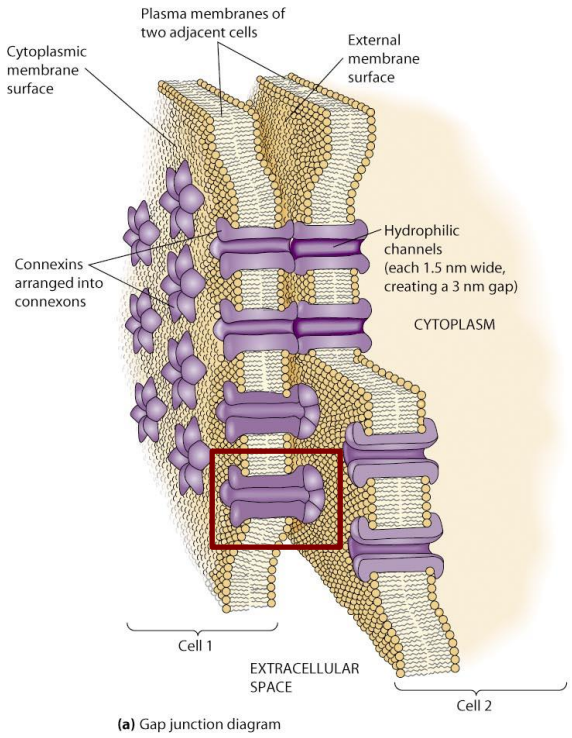
A protein on one cell membrane binds another prot on the adjacent cell to create gaps (pores) that connect cells

Small molecules and ions (ATP, cAMP, Na^+, Ca^{2+}, aa, vitamins)

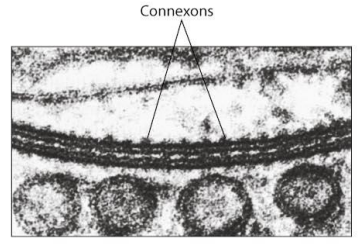
Signal **speed is very high** (e.g., heart, muscle, brain)

Connexons (hemichannels)

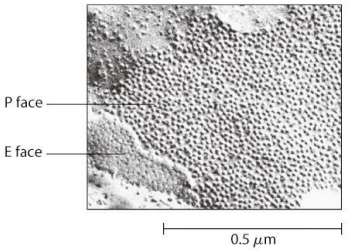
(<u>≤1.2 nM in diameter, 1 kDa of MW)</u>



(a) Gap junction diagram

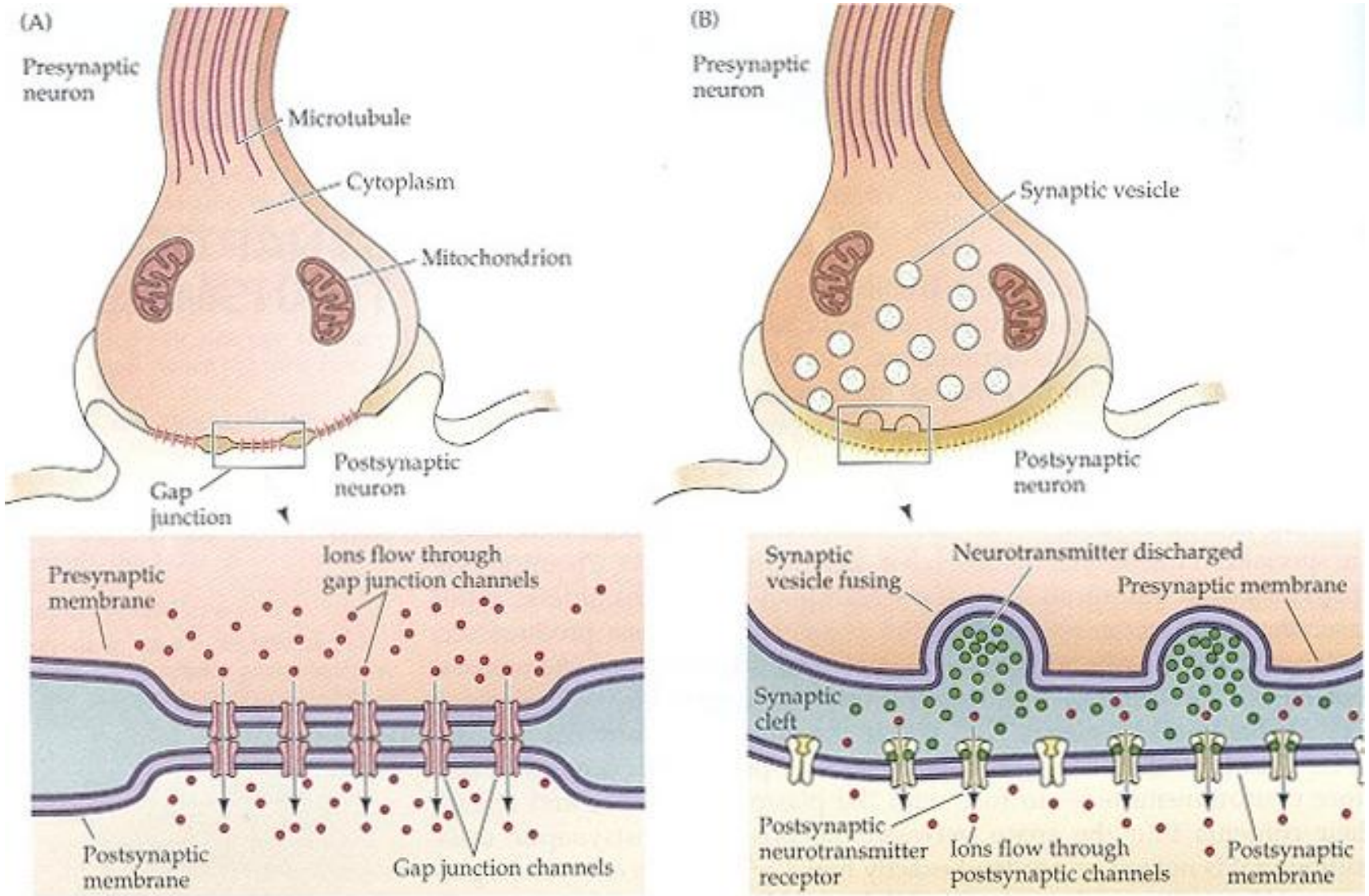


(b) Electron micrograph of a gap junction



(c) Freeze-fracture of a gap junction

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ELECTRICAL SYNAPSES

CHEMICAL SYNAPSES